

Ronit Shahu

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AI SCIENTIST | MACHINE LEARNING ENGINEER | NLP & GENAI

PROFESSIONAL SUMMARY

Data-driven AI professional with hands-on experience designing, deploying, and scaling production ML and NLP systems in regulated sectors (finance, healthcare). Proficient in Python, PyTorch, TensorFlow, scikit-learn, and LangChain, with expertise in model lifecycle management (MLflow, Azure ML, Databricks). Skilled in AI system design, explainable modeling, and business translation—partnering with stakeholders to deliver measurable impact. Experienced in containerized deployment (Docker, Kubernetes) and GenAI applications integrating LLMs, transformers, and reinforcement learning. Passionate about building ethical, interpretable, and scalable AI solutions that align with CIBC's trust and innovation values.

AREA OF EXPERTISE

Python	PyTorch & Tensorflow	Feature Engineering	Scikit – Learn
SQL	NLP / NLU	PySpark	AWS (EC2, Lambda)
Databricks	A/B Training / Testing	Pandas & Numpy	GIT / CI/CD / Jenkins / GitHub Actions
Kubernetes	RESTAPI – FastAPI / Flask	Azure ML	Generative AI (LLMs, LangChain, LangGraph, prompt engineering)

CAREER HIGHLIGHTS

AI-Driven Process Transformation: Delivered scalable, production-grade AI systems that reduced manual review time by 40%, improved data quality by 35%, and accelerated analytics pipelines across healthcare and financial workflows through automated data validation and model-driven insights.

End-to-End ML Systems Delivery & Leadership: Progressed from developer to technical lead, managing a 15+ cross-functional engineering team to deploy containerized ML APIs (FastAPI + Docker) with MLflow and CI/CD integration, enhancing model accuracy by 30% and enabling repeatable, compliant MLOps practices.

Continuous Improvement & Collaboration: Recognized as a Top Talent and Global AI Hackathon finalist for pioneering applications of LLM-based NLP and automated document intelligence, demonstrating a commitment to innovation, explainability, and cross-functional collaboration in applied AI delivery.

WORK EXPERIENCE

Lead Machine Learning Researcher, Canadian Robotics and Artificial Intelligence Ethical Design Lab, University of Ottawa, ON | Jan 2025 – May 2025

Appointed to lead a cross-functional team contributing to national AI ethics research.

- Led a 15-member data science team delivering ML-driven automation systems for national AI-ethics research, overseeing the full lifecycle from data engineering to model deployment.
- Collaborated with business analysts and risk teams to convert domain requirements into data-driven models and dashboards that informed product and policy decisions.
- Built and deployed containerized deep learning / ML APIs using FastAPI + Docker + GitHub actions, hosted on AWS EC2 and reproducible on Azure ML Workspaces with integrated MLflow experiment tracking, CloudWatch and CI/CD pipelines for versioned model delivery. Configured Linux-based model servers with Nginx reverse proxy and container networking security rules
- Designed feature pipelines and preprocessing workflows using Pandas / PySpark (Databricks-compatible) for real-time inference and analytics over structured & unstructured data.
- Deployed scalable microservices with automated scheduling and monitoring, improving cost efficiency by 50% and ensuring compliance with performance KPIs. Contributed to AI ethics research by developing transparent and auditable model reporting mechanisms aligned with explainable AI standards.

Software Developer – Life science and Health Care, Persistent Systems Limited, India | Dec 2021 – July 2023

- AI-powered Certification Verification System: Automated image & PDF validation pipeline using machine learning and OCR deployed via Azure Machine Learning pipelines, reducing manual verification by 70% and operational costs by 35%.
- Deployed and monitored ML models via Azure ML + Databricks pipelines, leveraging container orchestration and MLflow for lifecycle management and governance.
- Medical Data Extraction for Clinical Trials: Designed and trained custom spaCy NER models to extract cancer-related entities, integrated with Azure ML endpoints for deployment, reducing data preparation time by 40% and improving insight accuracy.
- Healthcare NLP Microservices: Developed and maintained Flask / FastAPI-based APIs containerized with Docker and orchestrated through Azure ML + CI/CD, improving information retrieval speed by 12% and enabling consistent model retraining. Familiar with generative architectures including Transformers, VAEs, and GANs for text and image synthesis tasks.
- Cross-functional Collaboration: Partnered with data scientists, DevOps, and clinical SMEs across global teams to translate healthcare data challenges into deployable, compliant ML solutions aligned with responsible AI practices. Knowledge of reinforcement learning and optimization methods for adaptive decision-making systems. Wrote advanced SQL and PySpark scripts for feature extraction and ETL across large structured datasets (>200k records) to support model training and evaluation.

EDUCATION

Master of System Science and Engineering (Interdisciplinary Artificial Intelligence) University of Ottawa, ON	2023 – 2025
Bachelor of Engineering in Electronics Engineering Shri Ramdeobaba College of Engineering and Management, India	2018 – 2022

AWARDS AND ACHIEVEMENTS

• Research Fellowship Scholarship, University of Ottawa, ON	2025
• Elected as Program Co-Ordinator for Systems Science & Eng. as a liaison between students and faculty.	2023
• Top Talent 2023, Persistent Systems (for measurable ML project impact), Persistent Systems Limited, India	2023
• 2nd Runner-Up, Global AI Hackathon Semicolons 2023, India	2023

CERTIFICATIONS

• In Progress: AWS Certified AI Practitioner	(Expected Q1 2026)
• Microsoft Azure AZ-900 Fundamentals	2022

PUBLICATIONS

• <u>Audio + Text based Emotion Recognition System</u>	2021
• <u>Detection of COVID-19 using Machine Learning and Android App Development</u>	2021

RELEVANT PROJECTS

Cancer-Aware Medical Chatbot with Research Recommender (*University of Ottawa, 2025*)

- Designed and deployed a **GenAI-orchestrated medical chatbot** using Dialogflow ES, FastAPI, and Google Gemini LLM—integrated with **AWS Bedrock** for evaluating multi-model responses (Claude, Titan) under secure, compliant governance. Combined TF-IDF, Doc2Vec, and K-Means similarity modules to recommend cancer research papers from Mayo Clinic datasets, optimizing latency by 45 % via prompt engineering and parallel API routing.
- Implemented modular LLM orchestration merging Gemini and Bedrock outputs with structured UMLS and Mayo Clinic APIs, demonstrating explainable retrieval-augmented generation and hybrid search workflows across clinical and research data sources. Applied NLP and NLU techniques (tokenization, entity linking, intent classification) to structure unstructured text and conversational data for LLM-based dialogue systems.

Credit-Risk Modeling & Explainable AI Pipeline (German Credit Dataset)

- Developed an end-to-end, cost-sensitive ML pipeline replicating enterprise credit-risk workflows using Python (scikit-learn, SHAP, MLflow). Engineered interpretable features with Weight-of-Evidence (WOE) / Information Value (IV) analysis, applied fairness audits, and encoded ordinal/nominal variables for Basel-aligned scorecards.
- Benchmarked Logistic Regression, Random Forest, and Gradient Boosting models (AUC ≈ 0.83) while minimizing expected misclassification cost (Bad → Good penalty 5×). Integrated SHAP explainability for global & local feature insights—ensuring transparency, bias detection, and audit-ready governance.